Advertising in Toxic Feeds: Ad Effectiveness & Brand Safety

Registered Report Stage 1 Protocol

Jonas Goergen[2] & Hauke Roggenkamp[1]

[1]ETH Zurich, Department of Management, Technology, and Economics
[2]Kedge Business School, Department of Marketing

Toxic content on social media captures attention and drives engagement, yet brands overwhelmingly treat it as a reputational threat and withdraw advertising from toxic environments. These two responses reflect two literatures that examine the effects of toxicity in isolation: brand safety research documents associative contamination from unsafe adjacent content, while platform engagement research documents attentional and behavioral gains from negative and toxic content. No existing work examines both within the same feed exposure. We integrate these perspectives in a single empirical framework. Using a dose-response design with stimulus sampling, we manipulate the proportion of toxic content in simulated social media feeds and measure feed-level engagement, ad-level attention, and brand-level reputational outcomes. We predict an inverted-U relationship between feed-level toxicity and both feed engagement and ad engagement: moderate toxicity elevates arousal and deepens processing; high toxicity triggers aversion and disengagement. The reputational effect of toxicity on brand attitude is theoretically ambiguous — the same attentional gains that benefit the ad also increase exposure to the aversive context — and is examined as a research question. Our design independently varies feed-level toxicity dose and adjacent-post toxicity, allowing simultaneous estimation of global and local effects. The findings inform brand safety strategy by providing a basis for evaluating the net effect of feed-level toxicity on advertising effectiveness.

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Words: 7658

"If it bleeds, it leads" - Eric Pooley

Online platforms have increasingly become home to toxic discourses. Harassment, hate speech, cyberbullying, identity attacks, and deliberately insulting language pervade feeds across topics and languages, driven by recommendation algorithms and virality mechanisms amplifying such content. This toxicity poses a challenge for the multi-billion dollar advertising industry operating on these platforms. When toxicity on X (formerly Twitter) escalated following changes in ownership and content moderation in late 2022, major advertisers including Apple, Disney, and IBM suspended their campaigns; the platform's ad revenue declined by more than half within a year (BBC News, 2023). This was not an isolated case. The World Bank pulled advertising from X in 2024 after a paid placement appeared alongside content from a white supremacist organization (CBS News, 2024). These incidents highlight a tension brands advertising online face: the same platforms that reach billions of consumers also expose branded content to environments brands cannot fully control. With global social media advertising projected to hit nearly $340 billion in 2026 (Statista 2026) and social channels attracting the largest share of incremental ad dollars worldwide (WARC 2025), the reputational and financial exposure and risk to brands is substantial.

Marketing research examines this risk to brands through the lens of brand safety. Recent work in this tradition documents that the proximity of branded content to unsafe or negative material can contaminate brand perceptions, even when no explicit link between the brand and the adjacent content exists (Grewal, Vana, and Stephen 2025). The general recommendation, followed by both theory and practice, is that any association with this type of content should be avoided. Accordingly, strategies securing brand safety, for example detection algorithms, blocklists, and moderation, have become a major point of concern for both brands and platforms. Among the unsafe content circulating online, toxicity represents a particularly prevalent and harmful source of contamination, given its harmful and targeted nature.

A separate stream of platform research studies the notion that toxic content can also have positive implications (Beknazar-Yuzbashev et al. 2022). It has been shown to drive attention and generate engagement. The platform-level implication seems different to the advice given to brands: instead of avoiding toxicity, platforms might have an interest in accepting toxicity to retain and engage users. From this perspective, feeds containing toxicity might expose co-located advertisements to larger and more engaged audiences. More attention is generally considered positive for an ad's effectiveness, but does that hold if this attention comes through and in a toxic context that might harm the brand. These two streams generate an interesting context for the same managerial question: Should my brand withdraw from toxic discourse or not? Brand safety research advises general withdrawal from toxic environments due to potential spillovers. Platform research implies that those same environments could

increase exposure, attention, and engagement. We argue that existing work cannot resolve this contradiction because each stream examines its mechanism in isolation and partly disregarding the naturalistic setting in which ads are experienced. Brand safety studies regularly measure ad and brand outcomes without accounting for the attention environment that toxicity creates. Platform studies measure platform-level outcomes without tracking whether certain spillovers might offset attentional or engagement gains. Marketing research has yet to examine both in combination and within the same feed exposure.

We address this gap by integrating platform dynamics, ad effectiveness, and brand safety within a single empirical framework. Focusing on digital advertising on social media, where toxicity is widespread and brands exercise little control over surrounding content, we examine whether the relationship between feed-level toxicity and brand advertising effectiveness and safety is more complex than either stream in isolation suggests. We manipulate the proportion of toxic content in simulated but more naturalistic social media feeds using a dose-response design (with stimulus sampling CITE) and measure ad effectiveness (attention and recall) as well as brand safety (attitudinal and behavioral) across toxicity levels.

Our central argument is that feed-level toxicity affects both ad effectiveness and brand safety outcomes through attentional mechanisms. At low to moderate levels, we expect feed toxicity to elevate attention. Elevated attention should affect ad effectiveness, measured in implicit dwell time and explicit recall, positively but brand safety, as attitudinal and behavioral responses to the brand, negatively; heightened emphasis on the brand in an increasingly toxic context harms said brand. At high toxicity, attention will be withdrawn from the branded ad, yielding decreased effectiveness but attenuated safety compared to moderate toxicity levels.

With this research, we intend to contribute to theory and practice on online advertising, brand safety, and platform design. First, we combine previously separate research programs on brand safety and online toxicity. Extending recent work on toxic behavior in gaming (Bernritter et al. 2025), we show that toxicity is an operationalizable construct with nuanced and unexpected effects for attention, ad effectiveness, and brand safety. Contrary to prior theorizing and practice, toxicity is not generally negative but might come with advantages - crucially conditional on dosage. Second, adding to brand and platform research, we demonstrate that both face a trade-off where toxicity is driving attention and advertising effectiveness, but at the potential expense of brand safety. Solving this tradeoff will depend on monitoring and moderating toxicity levels for their effect on multiple metrics across stakeholders. Third, we show that online research on advertising and platform dynamics can benefit from a more naturalistic empirical approach that combines implicit and explicit metrics of interest to different stakeholders with rigorous experimentation techniques. While controlled experiments can be helpful to isolate single variables and mechanisms, it may come at the cost of missing the varied effects of more natural interactions.

We proceed by presenting our theorizing around toxicity in online discourse and its effect on brand's advertising and safety, culminating in three hypotheses. After that, we provide motivating evidence for toxicity's effect on engagement from pilot data spanning XX A/B tests in the context of a news platform. Then, we lay out the envisioned experimental paradigm. This includes the design, procedure and stimuli, outcomes, along with an analysis plan and sample size justification. Further, we discuss pilot data that informed our empirical strategy and confirms its feasibility.

Theory

Toxicity in Online Discourse

Toxicity describes intentionally inflammatory communication directed at a target to provoke unproductive conflict (Hanscom et al. 2024). It subsumes harassment, hate speech, cyberbullying, identity attacks, profanity, and threats, unified by a common function of inflicting harm on a target (Hanscom et al. 2024; Lees et al. 2022; Shahi and Majchrzak 2025). Crucially, toxicity is a property of communicative style rather than subject matter as it characterises how a message is deployed irrespective of the topic it addresses. Toxicity's purpose is to damage rather than to evaluate, criticize, or persuade per se. Whereas constructive criticism, passionate disagreement, and even harsh evaluative language can be productive, toxicity has no regard for productive disagreement or advancing an argument.

Toxic discourse is pervasive across platforms, languages, and subjects (Avalle et al. 2024; Shahi and Majchrzak 2025). It emerges wherever users interact, not only around highly controversial subjects, such as politics, but also in entertainment, sports, and consumer contexts (Salminen et al. 2021; Tufa, Markov, and Possen 2024). Platform architectures amplify the prevalence of toxicity through recommendation algorithms, virality features, and network effects (Recuero 2024; Hanscom et al. 2024). The consequences of toxicity extend beyond the immediate online interaction and can translate to offline violence. For instance, recent evidence suggests that social media can act as a propagation mechanism for violent crimes by enabling the spread of extreme viewpoints (Müller and Schwarz 2021, 2023). Toxicity therefore generates regulatory pressure for platforms as legislation in the EU and elsewhere mandates moderation

of toxic content (Digital Services Act 2022; UK Online Safety Act 2023).

To meet these moderation demands, platforms can readily operationalize toxicity via automated classifiers (see e.g., Horta Ribeiro et al., 2023; Singhal et al., 2023). These assign continuous toxicity scores to individual posts, enabling moderation decisions that human manual review cannot produce at the volume of contemporary platforms (Jhaver et al. 2019). The most widely deployed toxicity classifier, Google's Perspective API, is trained on millions of human-annotated comments and runs on Reddit threads, The New York Times' comment sections, and Wikipedia, among other platforms (Lees et al. 2022; Perspective API 2024).

For brand safety practice, toxicity offers a complementary perspective. Because toxicity describes communicative style rather than subject matter, it differs from the content category-driven approaches that govern programmatic brand safety. Established industry standards and individual company solutions implement blocklists that classify content by subject matter: violence, adult material, hate speech, and similar categories (e.g., Amazon Ads, 2025; GARM, 2024). These operationalizations are coarse and produce both type-1 and type-2 errors. Keyword blocklists over-block safe inventory by flagging topical terms regardless of communicative intent: during the Euro 2024 final, keywords such as "shoot" or "attack" caused nearly half of a UK publisher's output to be excluded from advertising (The Media Leader, 2024), and as one advertising executive noted, the approach "frequently results in inventory that is contextually relevant, and safe, being blocked." Conversely, the same blocklists fail to detect unsafe content whenever a nominally harmless topic turns toxic through communicative style. Because toxicity is topic-independent, it addresses both failure modes simultaneously: it can flag unsafe contexts that topic-based filters miss while reducing exclusion of safe inventory. This captured dimension of communicative style is also what produces the attentional, attitudinal, and arousal effects relevant to advertising outcomes (see "Motivating Evidence" section).

Toxicity and Advertising

Toxicity is consequential for advertising along two dimensions. The first is ad effectiveness (i.e., whether the toxic context alters the attentional and mnemonic processing of the ad itself). The second is brand safety (i.e., whether the toxic context contaminates evaluations of the advertised brand). These dimensions are distinct yet interrelated. They capture different stages of the advertising process, attention and encoding on one hand and brand evaluation on the other, yet naturally brand harm requires prior exposure. An ad that wasn't noticed shouldn't harm or benefit the brand it carries. We develop hypotheses for each outcome in turn, before examining how attention connects them.

Ad Effectiveness

Evidence on toxicity and ad effectiveness is mixed. On the one hand, scholars find evidence for the claim from the infamous adage: "if it bleeds, it leads" (Pooley, 1989). Toxic content generates larger, wider, and deeper reply threads than civil counterparts on Twitter (Saveski et al. 2021) and across other platforms (Avalle et al. 2024), and reducing toxicity via a browser extension decreased user attention, time spent, impressions, and clicks (Beknazar-Yuzbashev et al. 2025). More generally, research on negative content demonstrated that it captures disproportionate attention (see, e.g., Section-Pilot and Pratto and John 1991; Rozin and Royzman 2001), receives higher engagement rates, greater sharing, and stronger recall than neutral and positive content (for an overview, see Robertson et al. 2023). Further, any threat-relevant stimuli, such as toxic communication (e.g., personal insult), activate automatic detection processes that elevate attention (Öhman and Mineka 2001)).

On the other hand, sustained exposure to severely toxic content produces aversion and withdrawal: users report avoiding participation for fear of harassment or shutting down accounts entirely, and self-censorship and silencing are among the observed reactions (Lenhart et al. 2016). Moderation studies find that reducing toxicity can increase participation [Matias et al. (2021); Wise et al. 2006], and widely applied operationalizations of toxicity, such as the Perspectives API mentioned before, explicitly frame a heightened propensity to leave a discussion as definitional (Lees et al. 2022; Aridor et al. 2024).

We argue that both effects may exist and that the level of toxicity represents the moderating variable, implying a non-monotonic relationship where ad effectiveness is increasing first and decreasing later. The cue-utilization hypothesis provides a potential unifying mechanism (Easterbrook 1959). At low to moderate toxicity, the provocative and targeted nature of toxic communication raises overall attention and pulls in users who engage by replying, retaliating, defending, or spectating. Importantly, the widened attentional window encompasses adjacent stimuli including the ad. At higher toxicity, attention is focused more on threat-relevant cues at the expense of peripheral stimuli (Martha and Sutherland 2014; van Steenbergen et al. 2011), and the ad — a commercially oriented, non-threatening element — loses attentional resources. Evidence from specific high-severity contexts is consistent with this: misogynistic hate speech reduces women's participation in political discourse (Carlson 2018), and toxicity decreases player engagement in gaming (Liu et al. 2025). The mixed evidence is thus not contradictory but reflects sampling across different positions on the toxicity continuum.

Ad effectiveness therefore depends on the toxicity dosage in the feed. At low-to-moderate doses, the attentional pull from toxicity exceeds aversion. At high doses, the relationship reverses. We therefore expect an inverted U: attention allocation increases with toxicity dose up to a threshold and declines beyond it.

> **H1:** Toxicity and ad attention follow an inverted-U relationship, such that attention increases with toxicity dose up to a point, beyond which it decreases.

Brand Safety

Prior literature suggests a negative relationship between feed toxicity and brand safety, where attitudinal or behavioral responses toward the brand are harmed as toxicity is increasing. Spillover contamination from adjacent unsafe content is a central mechanism explaining this pattern (Bernritter et al. 2025): consumers transfer negative associations from that content to the co-located brand, even absent any causal or semantic link between them (Grewal, Vana, and Stephen 2025) That is, when a brand is presented alongside unsafe content online, there is a spillover of the adjacent negativity that harms said brand. This spillover of contextual factors on co-located brands presented in the same context has also been demonstrated in settings beyond digital media, such as print and television (Bushman 2005; de Pelsmacker et al. 2002; Nian et al. 2021). Toxicity presents such an unsafe context as it is defined by harm inflicted on others. Following presented research, toxicity dosages should affect brand safety negatively.

> **H2:** Toxicity and brand attitudes follow a monotonically negative relationship, such that brand attitudes decrease as toxicity dose increases.

An ad that wasn't noticed shouldn't harm or benefit the brand it carries thus brand safety operates through attention. Only when enough attention is allocated to the ad in its toxic context, we would expect harm to spillover. For instance, Roggenkamp, Boegershausen, and Hildebrand (2025) found that there is no effect of unsafe content on brands among inattentive consumers. The core question then becomes how toxic content affects brand safety depending on attention.

Naturally scrolling through a social media feed distributes attention across dozens of competing stimuli. In prior research, participants regularly evaluate a curated and isolated pairing presented via static vignettes with minimal attentional competition (e.g., from a surrounding feed or earlier content exposure) – a condition that likely overstates the attention granted to any single adjacency in natural browsing. For example, Lee et al. (2021, e.g., Study 1) presented participants with a single offensive social media post on animal abuse with a pet shop brand ad presented above, alongside, and below the post.

Work on advertising has pointed out the relevance of testing advertisement effects in naturalistic settings (Eisend 2009) and highlighted the need to account for cluttered media environments with attention competition and dilution (Ha and McCann 2008). Translated to our context, brand safety should be threatened only when the brand-unsafe context cuts through the clutter and the pairing is processed as such; to test this we apply a more realistic environment where various content competes for attention (Roggenkamp, Boegershausen, and Hildebrand 2025).

We propose that under these conditions, the inverted U hypothesized for the effect of toxicity on attention under H1 will qualify the linear negative relationship under H2. Specifically, we expect that the attentional function linking toxicity to brand harm is itself non-monotonic in toxicity dose. For brand safety, low toxicity generates little attention and therefore little harm; the brand should stay safe; as toxicity is increasing to moderate levels, where attention is maximized, safety should be harmed the strongest; for high toxicity levels attention is partially withdrawn and even in the presence of high toxicity, the lack of attention attenuates damage to brand safety. Formally:

> **H3:** Toxicity and brand safety follow a non-linear relationship, such that brand safety decreases with toxicity dose up to a point, beyond which it partially recovers as attention withdraws.

The relationship between these three variables (attention, brand attitudes, and the resulting non-linear safety effect) is stylized in Figure 1.

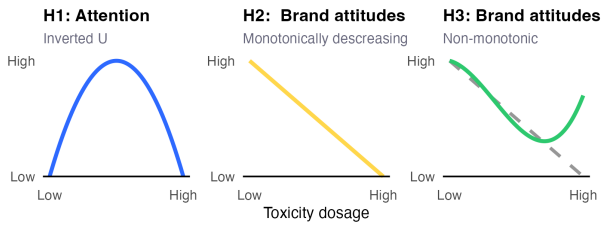

Figure 1. Stylized Hypothesized Relationships between Toxicity Dosage, Attention Allocation, and Brand Attitudes

Motivating Evidence

Research on social media platforms shows that toxic content drives *engagement*: it attracts more reactions, shares, and time spent than civil alternatives (Beknazar-Yuzbashev et al. 2022; Aridor et al. 2024; Beknazar-Yuzbashev, Jiménez-Durán, and Stalinski 2024). We provide evidence suggesting that toxicity also drives *attention* specifically, using 12,473 A/B tests from the *Upworthy Research Archive* (Matias et al. 2021) that cover 53,755 news headlines that were displayed 205.8M times.

In this setting, each A/B test randomly assigned website visitors to one of several headlines referring to the same article and recorded clicks on each headline. Unlike

likes or comments in a *social* media platform, these user reactions were not visible to others. The 2.8M clicks in this dataset therefore reflect attentional responses rather than social behavior, which is precisely the channel our theoretical argument concerns.

The recorded click-through rates (CTRs) range from .000% to 14.891%. The average CTR across all headlines is 1.394% and the median click rate is 1.067%. As the distribution among CTRs is right-skewed, only a small proportion of news stories are associated with a high CTR.

Toxic news headlines, defined as those exceeding a Perspective API score of 0.6, achieve higher click-through rates than civil alternatives ($Median_{CTR}^{Civil} = 1.061\%$; $Median_{CTR}^{Toxic} = 1.401\%$). Figure 2 provides model-free support for this claim: the distribution of click-through rates is shifted rightward for toxic headlines across the full range of observed values. A fixed-effects regression that compares headlines which refer to the same article quantifies the difference: the estimated uplift corresponds to 6.408% of the median click-through rate, and it holds after controlling for headline length, readability, and negative emotional valence.

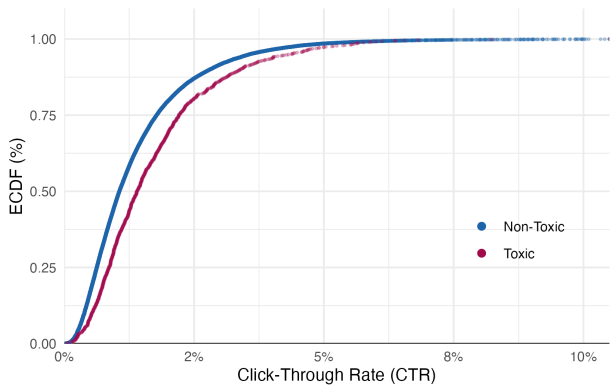

Figure 2. Empirical cumulative distribution functions of toxic vs. non-toxic headlines

Importantly, the Upworthy data represent an unusually clean source of field evidence for this purpose as the A/B tests were designed by the platform itself: visitors were randomly assigned to headline variants, one experiment ran at a time, and commercial incentives ensured careful implementation. Such data are usually proprietary and therefore rare. Even so, three limitations prevail. First, the data contain no information about the feed structure in which each headline was embedded, which is why we cannot derive a dose-response relationship between toxicity and attention. Furthermore, this study operationalizes clicks on the stimulus itself. It does therefore not offer evidence of excitation transfer from toxic content to adjacent, non-toxic content. Finally, it does not provide evidence on downstream consequences such as brand attitudes.

Addressing these limitations with additional field data is difficult, if not impossible. This is because field data from (social) media platforms are typically not available to researchers. Other data that stems from online platform studies and observational studies both suffer from algorithmic interference: platform algorithms determine which users see which content in ways that are unobservable to researchers. This makes it difficult to isolate the effect of toxicity on any outcome (Xu, Zhang, and Zhou 2020; Johnson 2023; Braun and Schwartz 2025). Furthermore, neither paradigm allows researchers to manipulate or even observe the context surrounding a focal ad, which is required to study spillover effects that characterize brand safety concerns.

Experimental Evidence

To address the three limitations of the news headline study, we designed a Digital In-Context Experiment to embed a sponsored post within a feed whose toxicity dose is experimentally varied. This enables a clean operationalization as well as more ecologically valid stimuli and measures compared to typical vignette experiments (Roggenkamp, Boegershausen, and Hildebrand 2025).

[Give a brief overview here]

Design

The study employs a dose-response design, which we implement with a Digital In-Context Experiment (*DICE*, see Roggenkamp, Boegershausen, and Hildebrand 2025; Chen, Schonger, and Wickens 2016). Specifically, we expose participants to a social media feed in which we manipulate the dosage of toxic content exogenously between participants. DICE renders the feed in a responsive browser interface that mimics the visual structure of a microblogging platform and records granular behavioral data unobtrusively (i.e., without requiring any explicit response from participants). Each participant views a single feed consisting of 25 organic posts around the movie *Avatar: Fire and Ash* (2025) and one sponsored post advertising snacks. The toxicity dose d, defined as the proportion of toxic organic posts, ranges from 0 (a fully civil feed) to 1 (a fully toxic feed) in increments of $\frac{1}{25}$. The resulting continuous manipulation allows estimation of the full dose-response function, to test the various functional forms predicted in H1-H3.

The construction or extraction of stimuli for a toxicity manipulation (e.g., via web scraping) is complicated because toxicity often correlates with several other content attributes. Manually retrieving and constructing matched pairs that differ only in toxicity is therefore difficult (if not infeasible), because any revision that removes toxic elements changes these latent dimensions as well. We address this issue by combining two methodological approaches: stimulus sampling (Simonsohn, Montealegre, and Evangelidis 2025) and scope testing with AI-gener-

ated stimuli (Tomaino et al. 2025). Accordingly, we created a corpus of 450 pre-constructed social media posts in which we vary the toxicity, other factors.

Posts in this corpus are classified as toxic (Perspective API score ≥ 0.6) or non-toxic (i.e., civil; score < 0.6), which yields two sampling pools. For a participant assigned dose d, the feed is assembled by sampling $d \times 25$ posts without replacement from the toxic pool and $(1 - d) \times 25$ posts from the civil pool, then randomly shuffling the resulting 25 organic posts. Each participant draws an independent sample, so participants assigned the same dose see different posts in a different order. Because idiosyncratic features of specific posts vary randomly across participants who are assigned the same dose, potential confounds are uncorrelated with the dose manipulation in expectation. (stimmt das überhaupt?)

We use generative AI to construct the corpus itself. This makes it tractable to generate posts across the full toxicity range while systematically varying orthogonal dimensions such as length, sentiment polarity, writing style, and thematic focus. The stimulus corpus was constructed in four stages. In the first stage, we prompted the Grok-3 API (xAI) to generate posts across four broad toxicity levels (i.e., civil, mildly toxic, toxic, and highly toxic) while independently varying length, writing style, sentiment, and thematic focus. Each post was scored using the Google Perspective API, which provides a continuous toxicity score on the $[0, 1]$ scale and constitutes the industry standard for computational toxicity detection in research and practice (Aridor et al. 2024). The initial stimuli generation produced adequate coverage at the extremes of the distribution but left a gap in the moderate range (scores between approximately 0.15 and 0.65). A second stage targeted this gap. Specifically, we applied a generate-and-filter procedure: posts were generated to three intermediate toxicity targets and accepted only if their toxicity score (predicted with the Perspective API) fell within the target band, otherwise generation was repeated. In a third stage, sampled posts were rephrased using another Grok-3 API call to eliminate repetitive phrasing patterns. Finally, we manually checked the corpus and edited few individual tweets. The resulting corpus (consisting of 450 organic posts) almost spans the full toxicity range [0.014, 0.939].

For each participant, we add the same sponsored post that advertises M&M's and invites them to enter a prize lottery by clicking on the post. The brand and product category were chosen to match the scenario framing: a snack brand is a contextually plausible advertiser in a feed about a movie. The sponsored post is inserted at a fixed position (14th scroll position) and is identical across all dose conditions.

Procedure & Measures

After providing informed consent, participants read a brief scenario framing the task: they had recently come across *Avatar: Fire and Ash* and were opening a microblogging feed to see what others were saying about the film. They are instructed to browse at whatever pace and depth feels natural and to click "Done browsing" when they feel they have seen enough. Afterwards, they are redirected to a Qualtrics survey automatically in where they first provide demographic information as a filler task. Next, we measure whether participants recall seeing the ad. Specifically, we measure free and aided recall. To measure brand recall, we show participants a list of 10 brands from different categories and ask them to indicate whether they recall seeing any of them (Campbell and Keller 2003; Simonov, Valletti, and Veiga 2025). Subsequently, participants rate the brand M&Ms on five items in an overall brand evaluation scale [Grewal, Vana, and Stephen (2025), e.g.,"I would consider buying products from the brand" with 1 = "strongly disagree," and 7 = "strongly agree"). They also complete a brand trust item ("I believe the brand is trustworthy"), in randomized order within the same block as the other items.

Participant behavior during feed exposure is recorded unobtrusively. A viewport tracker logs the duration each post is visible in the browser window (dwell time) to provide a proxy for (in)attention. Participants could also click on the ad (to sign up for an incentive compatible lottery), like posts and compose reply text. Total time from loading the feed in the browser to proceeding to the next page was recorded as overall time on feed. We also record paradata, such as the device type and screen resolution to allow for covariate adjustment for differences in feed rendering across devices.

Ad dwell time

This (log-transformed) variable is the primary outcome. It is the most direct unobtrusive measure of attention allocated to the sponsored post, recorded continuously by the viewport tracker without requiring any active response from participants. Dwell time shows high concurrent validity with mobile eye-tracking measures ($r = .89$ with gaze duration) and accounts for 76% of the variance in brand recall in validation data (Bruns et al. 2025). The sample size is powered for this outcome. H1 is supported or rejected on the basis of this outcome alone.

Recall

Unaided recall is a secondary outcome. Participants are asked to list any brands from which they recall having seen an ad in the feed. This captures whether exposure was sufficient to encode the brand into memory without a cue, and is stricter and less susceptible to guessing than aided recall. Aided recall, in which participants select from a list of eight brands the one they recall seeing a sponsored post for, is reported as a robustness check when unaided recall is sparse.

Lottery signup
This is binary variable. which we also consider as is a secondary outcome. Although behavioral and also a good approximation of financial brand safety (Bernritter et al. 2025), it is relatively sparse, which limits power. It is therefore reported for completeness but not treated as confirmatory.

Time on feed
Similar to the dwell time measure, this log-transformed outcome is reported as a feed-level variable. It addresses whether feed-level toxicity affects overall browsing duration. As such, the variable may predict (in)attention, but is not an ad effectiveness measure.

Brand attitude
is the primary outcome for H2. Participants rate agreement with five statements about M&M's on a 7-point scale (1 = *strongly disagree*, 7 = *strongly agree*); items are drawn from Grewal, Vana, and Stephen (2025) and cover purchase consideration, quality perception, recommendation likelihood, social identification, and general liking. The composite score is the mean across all five items. We apply the same two-lines procedure and decision rules as for H1 (see Table 2). If neither the two-lines test nor the quadratic term is significant, we conclude that feed-level toxicity does not affect brand attitude detectably.

Brand trustworthiness
Following Grewal, Vana, and Stephen (2025), brand trustworthiness is treated as a separate outcome rather than a component of the brand attitude composite. It is measured by a single item ("I believe M&M's is trustworthy") on the same 7-point scale and reported as a secondary outcome.

Pilot Data

Discuss correlation of recall and dwell time. Also discuss the scroll depth and that this results from the "stop browsing" button which we implemented to not create artificial dwell times when people just scroll to exit the feed. Mention that, for the main study, we will put the ad in the 10th position instead of the 14th as we have it in the pilot. This will result in a "loss" of X percent of the participants. Consider also analyzing how many participants recall seeing the ad who have not seen the ad. Then rewrite more or less everything and streamline it.

We recruited 201 participants for a pilot study. We operationalized feed toxicity differently for the pilot, compared to how we will operationalize it in the future. Specifically, we used a Gaussian kernel centered on target mean scores drawn from a uniform grid. Realized feed toxicity ranged from 0.105 to 0.842. The main study replaces this operationalization with the binary proportion method described in the Design section, which assigns toxicity dose as a fixed proportion of toxic posts and ensures uniform coverage across the full range.

The sponsored post entered the viewport of 136 participants (about two thirds of all participants). The remaining 65 did not scroll far enough for the ad to become visible. Among the 136 exposed participants, median ad dwell time was 1.936 seconds (mean 3.348s, SD 4.791s). On the log(1+s) scale used in the primary analysis, the pilot mean is 0.564 and SD is 1.204. These estimates ground the power analysis. Four participants spent fewer than 20 seconds browsing the feed.

Recall rates indicate adequate detection sensitivity. Seventy-two of 201 participants (35.8%) recalled M&M's on the unaided measure; 94 (46.8%) identified it on the aided measure. The gap between unaided and aided recall supports that both measures are informative and justifies treating aided recall as a robustness check. Lottery signup was sparse (6/201, 2.985%), consistent with the expectation in the Outcomes section. Brand attitude showed a pilot mean of 5.235 (SD 1.376) on the 1-7 scale, with no evidence of a ceiling effect.

Table 1. Summary statistics: Pilot data

Variable	N	Min	Max	Median	Mean	Sd
Ad dwell time	136	0.05	39.82	1.94	3.35	4.79
Time on feed	201	7.64	315.32	81.23	90.62	53.97
Scroll depth	201	1.00	26.00	17.00	17.58	7.03
Brand attitude	201	1.00	7.00	5.50	5.24	1.38
Recall unaided	201	0.00	1.00	0.00	0.36	0.48
Recall aided	201	0.00	1.00	0.00	0.47	0.50
Lottery signup	201	0.00	1.00	0.00	0.03	0.17

Analysis Plan

H1: Feed toxicity and ad engagement
H1 predicts that feed-level toxicity and ad engagement follow an inverted-U relationship.

All secondary outcomes are tested at $\alpha = .05$ using the same two-lines procedure as the primary outcome but are not the basis for supporting or rejecting H1. [We should rewrite this. This is intended to be a disclaimer that we will not correct for multiple hypothesis testing because the outcomes are secondary and because we only have on primary outcome per hypothesis (which are more or less unrelated).]

We follow the two-lines procedure proposed by Simonsohn (2018). For each outcome, we 1. fit a cubic spline to estimate the location of the peak, 2. apply the Robin Hood algorithm to determine the break point that maximizes power by allocating observations toward the weaker of the two slopes, and 3. estimate an interrupted regression with one slope below and one slope above the break point. H1 is fully supported if both slopes (denoted b_1 and b_2 in Table 2) pass one-sided tests ($\alpha = .05$) in the predicted directions: $b_1 > 0$ below the break point and $b_2 < 0$ above it. Because the two-lines test requires a true sign reversal, a positive-then-flat trajectory — in which engagement increases and then saturates without declining — would not satisfy the strict criterion but is

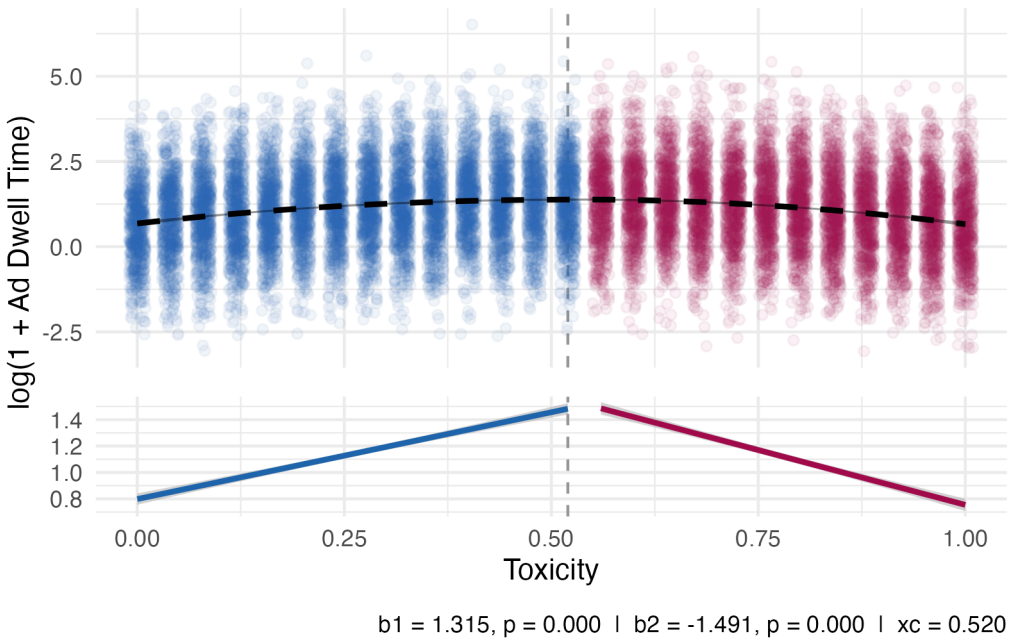

Figure 3. Two-lines test: feed toxicity and ad dwell time. Upper panel shows raw data with LOESS smoother; lower panel shows the two fitted regression lines. Break point determined by the Robin Hood procedure (Simonsohn 2018).

still broadly consistent with the arousal mechanism and is treated as partial support.

To capture this possibility, we additionally estimate a quadratic regression of the outcome on `feed_toxicity` and `feed_toxicity`^{2}. A significant negative quadratic coefficient alongside the plateau pattern in the two-lines test is interpreted as partial support for H1: the data show the predicted initial increase and a curvilinear compression, even if the second slope does not turn significantly negative. This secondary test is reported for completeness but, following Simonsohn (2018), is not treated as primary evidence given the known tendency of quadratic regression to produce false positives. The full decision rules are in Table 2.

Table 2. Pre-registered decision rules for H1. Quadratic term column is only evaluated when the two-lines test does not yield opposite significant slopes.

Two-lines result	Quadratic term	Interpretation
$b_1 > 0^*, b_2 < 0^*$	–	**Support H1** (inverted-U supported)
$b_1 > 0^*$, b_2 n.s.	$< 0^*$	**Partial support H1** (plateau: engagement increases and saturates but does not invert)
$b_1 > 0^*$, b_2 n.s.	n.s.	Reject H1 (monotonic increase)
$b_1 > 0^*, b_2 > 0^*$	–	Reject H1 (monotonic increase)
b_1 n.s. or b_2 n.s.	–	Reject H1 (no detectable relationship)

H2: Feed toxicity and brand attitude

H2 predicts an inverted-U relationship between feed-level toxicity and brand attitude, with contamination

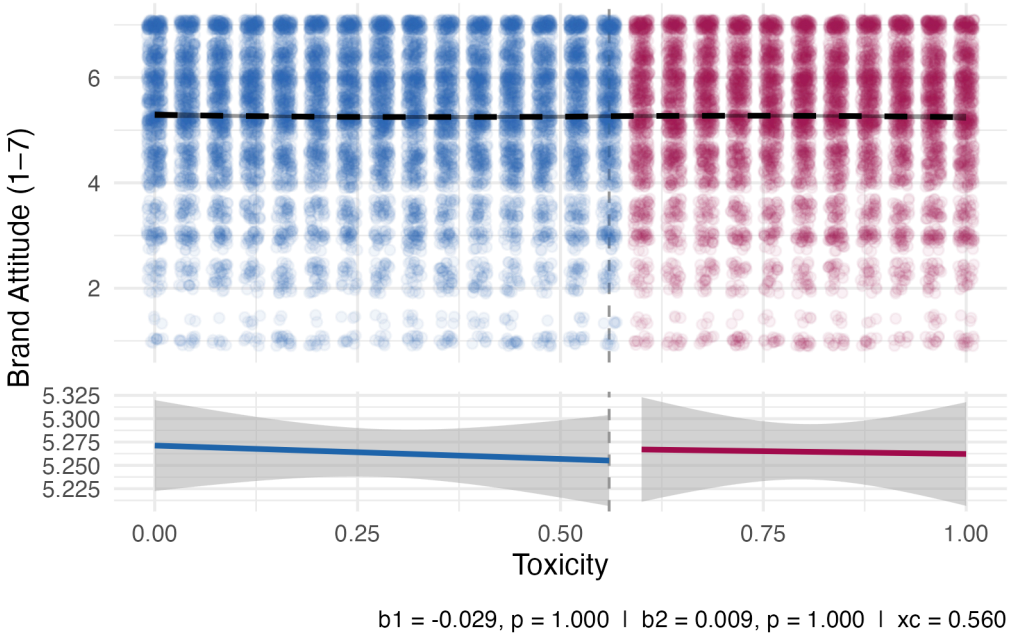

Figure 4. Two-lines test: feed toxicity and brand attitude.

strongest at moderate toxicity and attenuating at high toxicity. We apply the same two-lines procedure and quadratic fallback as for H1 (see Table 2). The primary test is whether both slopes are significant with opposite signs; partial support is declared if the first slope is positive and significant and the quadratic term is significantly negative.

We compute a composite brand attitude score as the mean of the five focal brand items (`gfb_1` to `gfb_5`), excluding the trustworthiness item (`gfb_6`) which is analysed separately. If neither the two-lines test nor the quadratic term is significant, we conclude that feed-level toxicity does not affect brand attitude in a detectable way, consistent with the attenuation argument in the brand safety section above.

Robustness and Potential Concerns

Because organic post order is randomized within each feed, the posts appearing immediately above and below the sponsored post vary across participants. The local toxicity of the ad's immediate context therefore co-varies with the feed-level dose in expectation but is not independently manipulated. To assess whether the estimated dose-response relationship is driven by local adjacency rather than overall feed composition, we re-estimate the primary analysis controlling for the mean Perspective API score of the two organic posts adjacent to the sponsored post (one above, one below). If the dose effect is attenuated once local adjacency is held constant, this would suggest that proximity to toxic content — rather than ambient feed-level toxicity — is the operative mechanism. Stability of the estimate would support the feed-level interpretation.

Sample Size Justification

The two-lines test has no closed-form power formula, so we estimate power by simulation following Lakens (2022). For each sample size $N \in [200, 400, \ldots, 4800, 5000]$, we

1. generate a dataset with two assumed inverted-U data-generating processes representing for small and medium effect sizes,
2. apply the two-lines test to the simulated data,
3. flag a success when both slopes pass a one-sided test ($\alpha = .05$; $b_1 > 0$ and $b_2 < 0$), and
4. repeat the procedure 1,000 times.

Power is defined as the probability that the test correctly rejects the null hypothesis when there actually is a genuine effect. In our simulation, power is therefore represented as the proportion of successful runs.

A key concept throughout is the inverted-U peak: under the assumed model $\mathbb{E}[Y \mid x] = \beta_0 + \beta_1 \cdot x(1-x)$, the function $x(1-x)$ reaches its maximum of 0.25 at $x = 0.5$ (a feed that is exactly half toxic). The peak outcome is therefore $\beta_0 + 0.25\,\beta_1$, and $0.25\,\beta_1$ is the peak uplift — how much higher the expected outcome is at 50% toxicity compared to a fully clean or fully toxic feed. Effect sizes are expressed as Cohen's d at this peak: $d = 0.25\,\beta_1/\sigma$, or equivalently $\beta_1 = 4\,d\,\sigma$.

We report power separately for two outcomes (ad dwell time and brand attitude) using this common d scale. The power curves (Figure 5) indicate that detecting a small effect ($d = 0.2$) with 80% power requiresapproximately

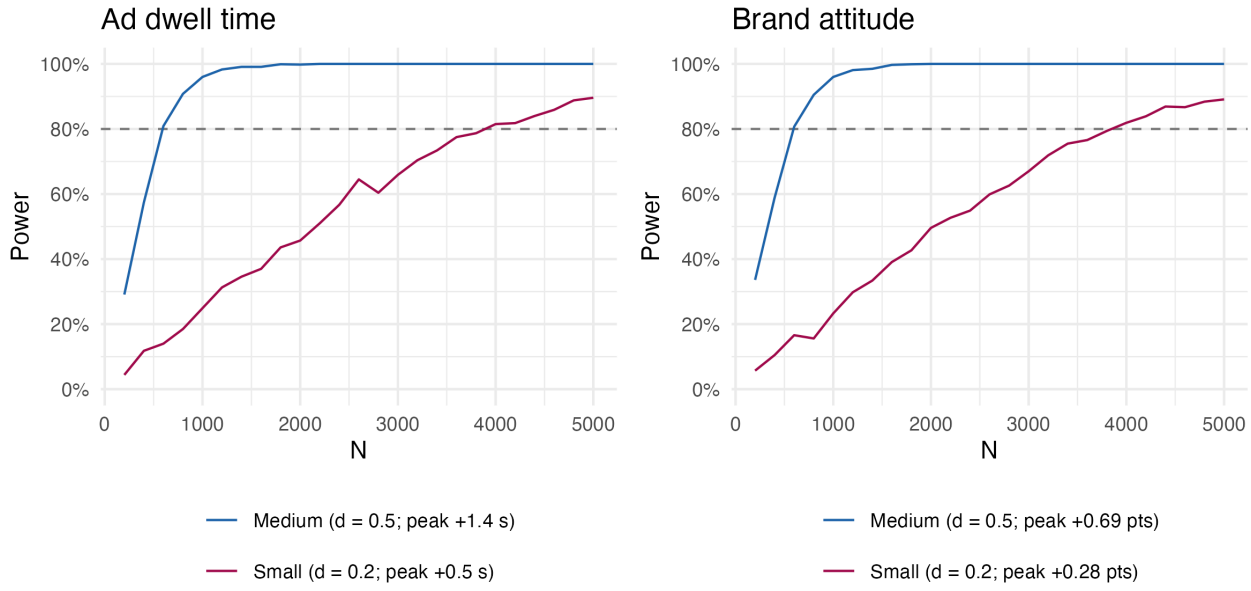

Figure 5. Power curves for the two-lines test on ad dwell time (left) and brand attitude (right). Effect sizes are expressed as Cohen's d at the inverted-U peak (d = peak uplift / pilot SD). Dashed line marks 80% power.

$N = 4,000$ participants; we adopt this as our recruitment target. The two-lines test is more conservative than a simple quadratic regression, so this sample size provides at minimum adequate power for the pre-registered primary test.

Ad dwell time. Because ad dwell time is heavily right-skewed, we analyze it on the $\log(1 + \text{seconds})$ scale, with σ estimated from the pilot. The pilot baseline is 0.8 s (median dwell time at the tails, i.e., in a fully clean or fully toxic feed). The peak — a 50% toxic feed — raises median dwell time to 2.2 s under a medium effect ($d = 0.5$), an absolute increase of 1.5 s. Under a small effect ($d = 0.2$), the peak median is 1.2 s.

Brand attitude. We apply the same d = 0.5 / d = 0.2 convention. The pilot mean is 5.24 on a 1–7 scale (SD: 1.38). A medium effect raises peak-condition attitude by 0.69 points; a small effect by 0.28 points.

Recruitment

Exclusion Criteria

References

Aridor, Guy, Rafael Jiménez-Durán, Ro'ee Levy, and Lena Song. 2024. "The Economics of Social Media." *Journal of Economic Literature* 62 (4): 1422–74. https://doi.org/10.1257/jel.20241743.

Beknazar-Yuzbashev, George, Rafael Jiménez Durán, Jesse McCrosky, and Mateusz Stalinski. 2022. "Toxic Content and User Engagement on Social Media: Evidence from a Field Experiment." Working Paper, November. https://doi.org/10.2139/ssrn.4307346.

Beknazar-Yuzbashev, George, Rafael Jiménez-Durán, and Mateusz Stalinski. 2024. "A Model of Harmful yet Engaging Content on Social Media." *AEA Papers and Proceedings* 114 (May): 678–83. https://doi.org/10.1257/pandp.20241004.

Bernritter, Stefan F., Ilias Danatzis, Jana Möller-Herm, and Francesca Sotgiu. 2025. "Navigating Toxic Playgrounds: Managing Reputational and Financial Brand Safety in Multiplayer Video Games." *International Journal of Research in Marketing* 42 (4, Part A): 936–56. https://doi.org/10.1016/j.ijresmar.2025.01.002.

Braun, Michael, and Eric M. Schwartz. 2025. "Where a/b Testing Goes Wrong: How Divergent Delivery Affects What Online Experiments Cannot (and Can) Tell You about How Customers Respond to Advertising." *Journal of Marketing* 89 (2): 71–95. https://doi.org/10.1177/00222429241275886.

Bruns, Daniel, Julian Felix Kopka, Lennart Borgmann, Steffen Prior, and Tobias Langner. 2025. "Measuring Gaining and Holding Attention to Social Media Ads with Viewport Logging: A Validation Study Using Mobile Eye-Tracking." *Journal of Advertising* 54 (5): 655–72. https://doi.org/10.1080/00913367.2025.2524186.

Campbell, Margaret C., and Kevin Lane Keller. 2003. "Brand Familiarity and Advertising Repetition Effects." *Journal of Consumer Research* 30 (2): 292–304. https://doi.org/10.1086/376800.

Chen, Daniel L., Martin Schonger, and Chris Wickens. 2016. "oTree—an Open-Source Platform for Laboratory, Online, and Field Experiments." *Journal of Behavioral and Experimental Finance* 9: 88–97. https://doi.org/10.1016/j.jbef.2015.12.001.

Grewal, Lauren S., Prasad Vana, and Andrew T. Stephen. 2025. "Brands in Unsafe Places: Effects of Brand Safety Incidents on Brand Outcomes." *Journal of Marketing Research* 62 (6): 981–1002. https://doi.org/10.1177/00222437251349522.

Johnson, Garrett A. 2023. "Inferno: A Guide to Field Experiments in Online Display Advertising." Journal Article. *Journal of Economics & Management Strategy* 32 (3): 469–90. https://doi.org/10.1111/jems.12513.

Lakens, Daniël. 2022. "Improving Your Statistical Inferences." Zenodo. https://doi.org/10.5281/zenodo.6409077.

Lees, Alyssa, Vinh Q. Tran, Yi Tay, Jeffrey Sorensen, Jai Gupta, Donald Metzler, and Lucy Vasserman. 2022. "A New Generation of Perspective API: Efficient Multilingual Character-Level Transformers." In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 3197–207. KDD '22. New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3534678.3539147.

Matias, J. Nathan, Kevin Munger, Marianne Aubin Le Quere, and Charles Ebersole. 2021. "The Upworthy Research Archive, a Time Series of 32,487 Experiments in u.s. Media." *Scientific Data* 8: 195. https://doi.org/10.1038/s41597-021-00934-7.

Öhman, Arne, and Susan Mineka. 2001. "Fears, Phobias, and Preparedness: Toward an Evolved Module of Fear and Fear Learning." *Psychological Review* 108 (3): 483–522. https://doi.org/10.1037/0033-295X.108.3.483.

Robertson, Claire E., Nicolas Pröllochs, Kaoru Schwarzenegger, Philip Pärnamets, Jay J. Van Bavel, and Stefan Feuerriegel. 2023. "Negativity Drives Online News Consumption." *Nature Human Behaviour* 7: 812–22. https://doi.org/10.1038/s41562-023-01538-4.

Roggenkamp, Hauke, Johannes Boegershausen, and Christian Hildebrand. 2025. "DICE: Advancing Social Media Research Through Digital in-Context Experiments." *Journal of Marketing*. https://doi.org/10.1177/00222429251371702.

Simonov, Andrey, Tommaso Valletti, and Andre Veiga. 2025. "Attention Spillovers from News to Ads: Evidence from an Eye-Tracking Experiment." *Journal of Marketing Research* 62 (2): 294–315. https://doi.org/10.1177/00222437241256900.

Simonsohn, Uri. 2018. "Two Lines: A Valid Alternative to the Invalid Testing of u-Shaped Relationships with Quadratic Regressions." *Advances in Methods and Practices in Psychological Science* 1 (4): 538–55. https://doi.org/10.1177/2515245918805755.

Simonsohn, Uri, Andres Montealegre, and Ioannis Evangelidis. 2025. "Stimulus Sampling Reimagined: Designing Experiments with Mix-and-Match, Analyzing Results with Stimulus Plots." *Journal of Personality and Social Psychology* 129 (1): 71–90. https://doi.org/10.1037/pspa0000449.

Tomaino, Geoff, Asaf Mazar, Ziv Carmon, and Klaus Wertenbroch. 2025. "A Simple Method for Improving Generalizability in Behavioral Science: Scope Testing with AI-Generated Stimuli (STAGS)." *Consumer Psychology Review* 8 (1): 87–97. https://doi.org/10.1002/arcp.1101.

Xu, Heng, Nan Zhang, and Le Zhou. 2020. "Validity Concerns in Research Using Organic Data." *Journal of Management* 46 (7): 1257–74. https://doi.org/10.1177/0149206319862027.

Stimulus Corpus Construction

Constructing a stimulus corpus that spans the full toxicity range while holding other content attributes constant is not feasible with scraped content. Manually matching or adjusting scraped posts to isolate toxicity from latent attributes is both labor-intensive and practically limited. Manual revision that removes toxic language may also changes tone, length, and affect. We therefore generate the corpus synthetically using a large language model (see also Tomaino et al. 2025), which allows systematic variation of toxicity level on the post level.

We address concerns ecological validity through two design choices. First, we calibrate all generated posts against the Google Perspective API (Lees et al. 2022). Second, we rephrase all posts through a separate humanization step that eliminates repetitive LLM phrasing patterns and introduces realistic orthographic variation (e.g., missing apostrophes, dropped word-endings) calibrated to the level present in organic social media text. We anchored the humanization prompt by a set of real posts scraped from the social media platform X about *Avatar: Fire and Ash*, which serve as few-shot register examples. These design choices are intended to ensure that the posts read as authentic organic discourse rather than machine-generated text, while preserving the controlled toxicity distribution that natural data cannot provide.

Generation Pipeline

We constructed the corpus in four stages: (1) initial generation across the full toxicity spectrum, (2) gap-filling via a generate-and-filter loop, (3) humanization and toxicity rescoring, and (4) manual review.

Stage 1: Initial generation. We prompted the Grok-3 API (xAI) to generate posts about *Avatar: Fire and Ash* (2025) across four broad toxicity levels: civil, mildly toxic, toxic, and highly toxic. Each generation call independently sampled from a set of orthogonal variation dimensions – post length (very short to long), writing style (casual, formal, emphatic, abbreviated, deadpan, meme-register), thematic focus (specific characters, box office performance, visual effects, personal reactions, comparisons to earlier entries in the franchise), and post format (statement, rhetorical question, reaction, listicle, fan theory). A factual context block injected into the system message grounded all posts in verified information about the film (cast, characters, box office gross, critical reception, Cameron's stated plans for sequels). A discourse-hook dimension sampled from a list of recurring fan debate topics (e.g., the underuse of the villain Varang, the divisive portrayal of Spider, Disney's reported plans to reduce future budgets) to ensure posts reflected the actual topical structure of audience discourse rather than generic movie reactions.

Stage 2: Gap-filling. Initial generation produced adequate coverage at the low and high extremes but left sparse coverage in the moderate range (Perspective API scores approximately 0.15-0.65) and in the extreme upper tail (scores above 0.80). Two supplementary scripts addressed these gaps using a generate-and-filter loop: posts were generated with target-band guidance and accepted only when their Perspective API score fell within the specified interval; otherwise generation was retried up to a fixed maximum. The moderate-range script targeted three sub-bands (0.15-0.35, 0.35-0.50, 0.50-0.65). The extreme-range script crossed two upper bands (0.65-0.80, 0.80-1.00) with five content moderation categories (harassment, hate speech, threatening language, sexually explicit content, and content promoting illicit activity, following the OpenAI moderation taxonomy), ensuring variety within the toxic pool beyond simple profanity. A fourth script targeted the low end of the civil range (0.00-0.15), covering fan enthusiasm, craft appreciation, and emotional responses that are absent from generation pipelines optimized for toxicity variation. All generation scripts used the same factual context block, discourse hooks, and variation dimensions as Stage 1.

Stage 3: Humanization and rescoring. All generated posts were pooled and sampled to approximate a uniform distribution across ten equal-width toxicity bins (target: 50 posts per bin; 500 posts total). Because humanization systematically shifts Perspective API scores downward (rephrased posts were generally less aggressive in surface form than the originals) the three highest bins were oversampled before humanization (by factors of 2, 3, and 5 respectively) and re-capped at the target after rescoring. Sampled posts were sent in batches to Grok-3 with instructions to rephrase each post so it reads as a different real person's contribution, while preserving the toxicity intensity indicated by its Perspective API score. The rephrasing prompt included a structural avoidance list (overused openings and trigrams detected by n-gram frequency analysis across the sampled pool), a set of nine register-calibrated few-shot examples spanning the full toxicity range, and instructions varying post length, movie reference style, and thematic angle. All posts were then rescored with the Perspective API, and the final corpus was assembled by re-capping at 50 posts per bin on the rescored distribution. Typos were injected post-hoc.

Stage 4: Manual review. We then read through the full corpus and identified repetitive phrasing, implausible word choices, and other linguistic artifacts of generation. Edits were minimal and the most common correction was replacing "flick" with "movie" or "film", which appeared with unnatural frequency.

Corpus Characteristics

The final corpus contains 450 organic posts spanning a Perspective toxicity range of [0.014, 0.939] (mean = 0.454, SD = 0.265). Applying the classification threshold of 0.6

yields 298 civil posts (66.2%) and 152 toxic posts (33.8%), which yields sampling pools of sufficient depth for the dose-response design.

Potential confounds. A concern with the constructed stimulus corpus is that the intended manipulation — toxicity — may co-vary with unintended text features that could independently affect participant responses. Two plausible confounds are post length and readability. Word count showed no systematic relationship with toxicity (Spearman ρ = -.10), confirming that longer posts are not disproportionately concentrated in any part of the toxicity range. Readability (Flesch-Kincaid grade level) was likewise approximately uniform across bins (see Table 3). Beyond these surface features, all posts share a common topic (Avatar: Fire and Ash), were generated from the same prompt structure, and cover a comparable range of aspects (plot, cast, visuals, comparisons to prior instalments).

Table 3. Post length and readability by toxicity bin. Word count and Flesch-Kincaid grade level (FK-GL) are averaged within each bin. Absence of a monotone trend indicates that toxicity does not co-vary with these surface features.

Bin	N	Words M	Words SD	FK-GL M	FK-GL SD
[0–0.1)	50	34.3	15.9	15.1	1.8
[0.1–0.2)	50	29.6	12.2	15.7	2.2
[0.2–0.3)	50	32.1	13.5	15.1	2.3
[0.3–0.4)	50	30.2	14.5	15.9	3.0
[0.4–0.5)	48	32.7	12.3	16.3	2.2
[0.5–0.6)	50	32.6	12.1	16.3	2.2
[0.6–0.7)	50	32.1	11.2	16.6	2.9
[0.7–0.8)	50	29.1	12.0	16.4	2.1
[0.8–0.9)	41	25.9	10.9	15.7	2.6
[0.9–1)	11	27.0	8.7	15.8	1.6

Human Validation of Toxicity Scores

The Perspective API score operationalizes toxicity dose in the main experiment, so establishing its criterion validity matters. We validate it against two independent sources: 41 Prolific participants who rated a stratified subset of the stimulus corpus, and GPT-4o-mini (version `gpt-4o-mini-2024-07-18`), which scored every post in the full corpus using the same construct definition as the human raters.

Design

We drew 100 posts from the stimulus corpus stratified across the ten Perspective score bins (10 per bin). We recruited participants on Prolific, restricting the sample to Canadian and US-American residents with a task-approval rate of at least 99% and self-reported English fluency.

The study ran in oTree (Chen, Schonger, and Wickens 2016). Before rating, participants read the following toxicity definition and then worked through three annotated calibration examples (one civil, one borderline, one clearly toxic) to anchor their judgments:

> *"language that a reasonable person would find attacking, devaluing, or harmful – or that disrupts civil, constructive discourse."*

Each participant then rated a random selection of 36 posts on a continuous slider (0 = "Not at all toxic", 100 = "Extremely toxic"), six per page, with the definition visible as a header throughout. Two attention checks were interspersed at fixed positions (one unambiguously civil post and one unambiguously toxic post) and participants who rated the civil check above 40 or the toxic check below 60 were excluded.

Analysis

The Perspective API score is designed to predict the probability that a given reader would find a comment toxic: a score of 0.7 is intended to mean that roughly 70% of readers would consider the post toxic. We mirror this operationalization and calculate the proportion of raters who assigned a score above 0.6 (on the 0–1 scale). We then compute three Pearson correlations: the Perspective API score against the mean human rating per post, the Perspective API score against the proportion of raters who exceeded 0.6, and the Perspective API score against the GPT-4o-mini toxicity score across the full corpus (see panels A, B, and C, respectively in Figure 6). The first two correlations describe criterion validity against human judgment via complementary aggregation schemes whereas the third provides a model-independent cross-check that requires no human data.

Results

All three panels support using the Perspective API score as a reliable measure of post-level toxicity. Mean human ratings tracked Perspective API scores closely ($r = .888$, 95% CI [.838, .924], $t(98) = 19.15$, $p < .001$), and agreement was essentially unchanged when the continuous mean was replaced by the proportion-based operationalisation ($r = .895$, 95% CI [.847, .928], $t(98) = 19.85$, $p < .001$). This suggests that the finding does not depend on the aggregation choice. Human raters and GPT-4o-mini also agreed within the validation subsample ($r = .852$, 95% CI [.784, .900], $t(90) = 15.45$, $p < .001$). Panel C extends this to the full corpus, where the two automated classifiers, which differ in architecture, training data, and scoring rubric, aligned tightly ($r = .804$, 95% CI [.765, .837], $t(374) = 26.13$, $p < .001$). The convergence of human raters, a purpose-built toxicity classifier, and a general-purpose language model on the same posts provides evidence that the Perspective API score captures a coherent, human-interpretable dimension of post toxicity.

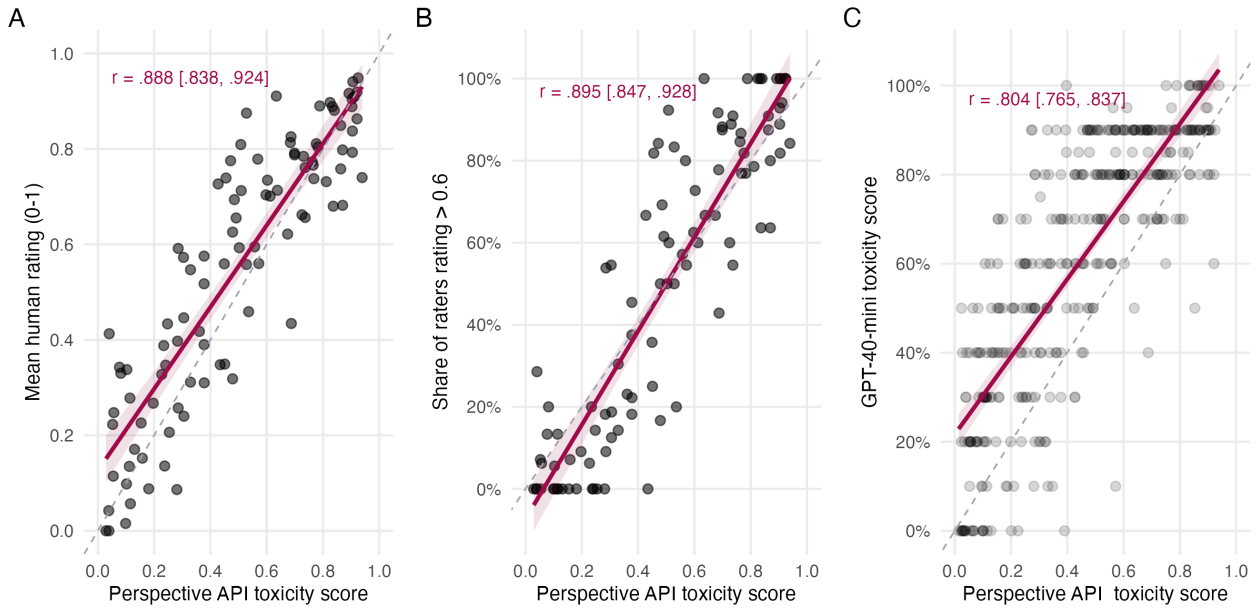

Figure 6. Convergent validity of automated toxicity measures. Panels A and B are based on the validation subsample (*n* = 100 posts, each rated by an average of 12 Prolific participants). Panel A shows the correlation between mean human ratings (0–1 scale) and the Perspective API score; Panel B shows the same Perspective API score against the share of raters who rated the post above 0.6. Panel C is based on the full synthetic corpus and shows the correlation between the Perspective API score and the GPT-4o-mini toxicity score.